

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF PHYSICS
MIDSESSION EXAMINATION

PHYS3114 – Electrodynamics
Term 3, 2020

1. Time allowed – 50 minutes
2. Total number of questions – 2
3. Total marks available – 20
4. Answer ALL questions
5. QUESTIONS ARE NOT OF EQUAL VALUE.
6. Answers should be uploaded in electronic format to the Moodle portal at the conclusion of the exam.
7. Additional time will be allocated for submitting your answers
8. **Candidates must verify the upload is successful and contact the exam convener immediately if there is an issue.**
9. Answer all questions in a single book.
10. University-approved calculators may be used.
11. This paper may be retained by the candidate.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_g}{D} x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D} y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (3)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (4)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (5)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (6)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (7)$$

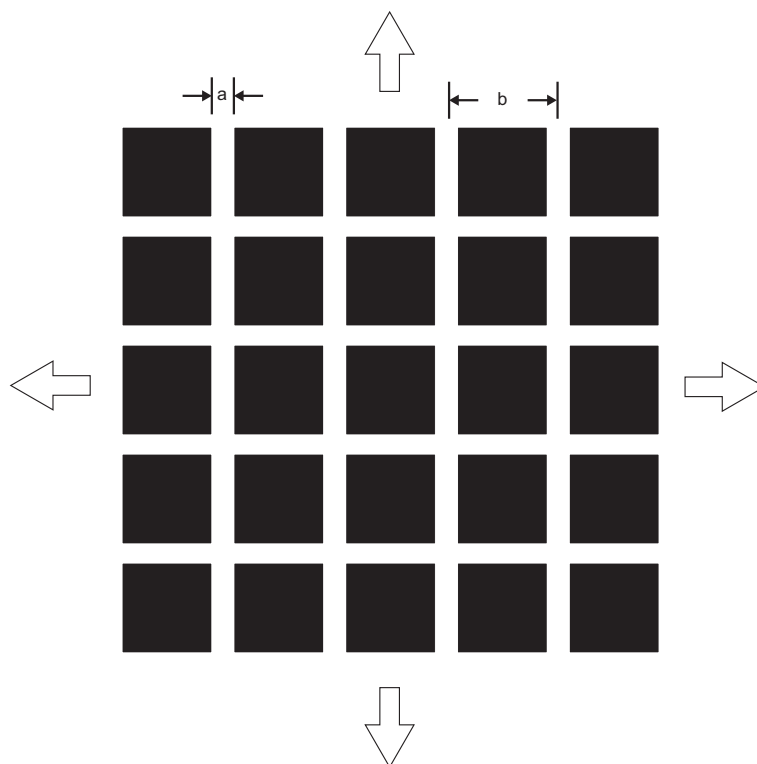
$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (8)$$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

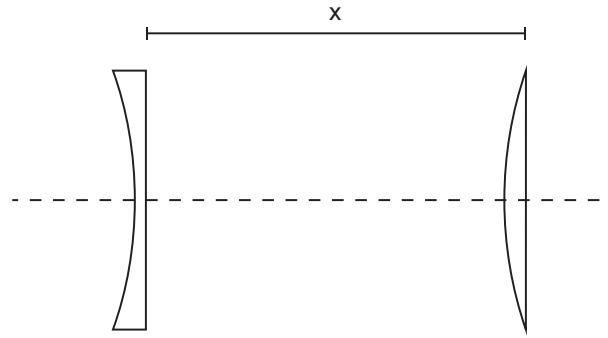
Question 1 (12 marks)

Consider the following aperture to be used in a Fourier filtering application. It consists of a 2D square grid of transparent intersecting vertical and horizontal lines of width, a , that are separated by a spacing, b . The pattern extends in both directions over a large number of periods. The aperture is placed in the object plane of an optical diffractometer and illuminated with a uniform plane wave of wavelength, λ .



- Construct an expression for the aperture, $f(x, y)$, in terms of common Fourier functions.
- With the aid of appropriate Fourier theorems, derive an expression for the diffracted field, $F(u, v)$, observed at the back focal plane of the transform lens.
- Determine an expression for a Fourier filter, $T_1(u, v)$, that would just allow the diffraction peaks along the u axis to be transmitted to the image plane of the optical diffractometer.
- Derive an expression for the intensity pattern that would be observed in the image plane after filtering the diffraction pattern with $T_1(u, v)$. Sketch the result.
- The Fourier filter is translated in the v direction such that it coincides with the next set of diffraction peaks. Calculate the resulting intensity pattern in the image plane and comment on the result.
- The filter is again placed at the origin and the width of the filter is increased such that successive diffraction orders in the v direction are transmitted. How will the filtered image change?

Question 2 (8 marks)



An optical system contains two lenses separated by a distance x . The first is a planar-concave lens (L1) a radius of curvature of $|R_1| = 10$ cm. A second lens (L2) is a planar-convex lens with a radius of curvature of $|R_2| = 30$ cm. Both lenses are composed of a glass with refractive index of $n = 1.5$. [hint: don't forget the sign conventions]

- (a) calculate the lens separation, x , such that a collimated light beam entering the first lens will be collimated at the output.
- (b) if the width of the collimated beam entering the optical system is 5mm, use the matrix method to determine the width of the exiting beam?
- (c) Does a concave lens do a Fourier transform during image formation? Use an analysis of the phase response of the lens to justify your answer.